

Singapore AI services — GPU hub positioning and enterprise adoption

Three hyperscalers have committed **over USD 10 b to Singapore GPU-grade infrastructure through 2028**, while the Model AI Governance Framework v2 shapes enterprise deployment. Power and talent — not demand — are the binding constraints on the city-state's regional hub thesis. Strategic outlook through 2031.

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COUNTRY

Singapore (SG)

INDUSTRY

AI Services & Infrastructure

SCOPE & HORIZON

2027 5y forecast

REPORT ID

KR-SGP-AISVC-2027-001

How we built this view

This view is built from two evidence streams: **primary inputs** — operator and enterprise channel reads on compute capacity, LLM deployment timelines, and governance posture — and **secondary anchors** — IMDA, EDB, and MAS publications, hyperscaler investment disclosures, and triangulated market estimates.

PRIMARY INPUTS

Operator & enterprise reads	Reads on GPU compute capacity build-out, enterprise LLM deployment timelines and managed-service adoption drawn from hyperscaler investment releases, enterprise software filings, and sector practitioner commentary from finance, logistics, and government verticals.
Analyst synthesis	KIRA in-house triangulation across substantive English-language research queries, pattern-matching against regional digital-infrastructure and AI-services studies. Tagged [Kira estimates] throughout.
Governance reads	Analysis of IMDA and PDPC publications including Model AI Governance Framework v2, the National AI Strategy 2.0, and MAS guidance on responsible financial-sector AI. Calibrated against implementation timelines.

SECONDARY ANCHORS

Statistical authorities	Infocomm Media Development Authority (IMDA), Economic Development Board (EDB), Ministry of Finance (MOF), and Singapore Department of Statistics. Source of digital-economy, investment-approval, AI-budget, and employment anchors, cited inline.
Hyperscaler filings	AWS, Google Cloud, and Microsoft Azure Singapore investment announcements and region-expansion disclosures. NVIDIA DGX Cloud partnership terms and NUS/NTU research infrastructure commitments used as capital proxy metrics.
Triangulated estimates	Enterprise adoption rates and AI-services spend splits where single-source disclosure is thin. We combine 2-3 inputs and disclose method. Tagged [Kira estimates] with anchor metric noted inline. See endnote.

Contents

Ten sections across fourteen pages, with dark dividers marking the macro, infrastructure, enterprise-demand, and governance cuts. Forecast and methodology endnote close the document.

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Singapore AI services — 2027 inflection

An AI-services market estimated at **USD 5.4 b in 2026** and growing above 28% annually, anchored by over USD 10 b in hyperscaler compute commitments, a Model AI Governance Framework v2 that codifies responsible deployment, and an enterprise sector moving from pilots to production. **Power, talent, and governance compliance** now separate leaders from followers.

AI SERVICES MARKET 2026

USD 5.4_b

~28% CAGR to 2031 [Kira estimates]

[KIRA EST]

HYPERSCALER COMPUTE CAPEX

USD 10.3_b

AWS + Google + Microsoft committed [EDB 2026]

[EDB 2026]

AI STARTUPS IN ECOSYSTEM

~1,100

+22% YoY; 180+ with Series A+ [IMDA 2026]

[IMDA 2026]

GOVERNMENT AI BUDGET 2026

SGD 1.0_b

National AI Strategy 2.0 multi-year envelope [MOF 2026]

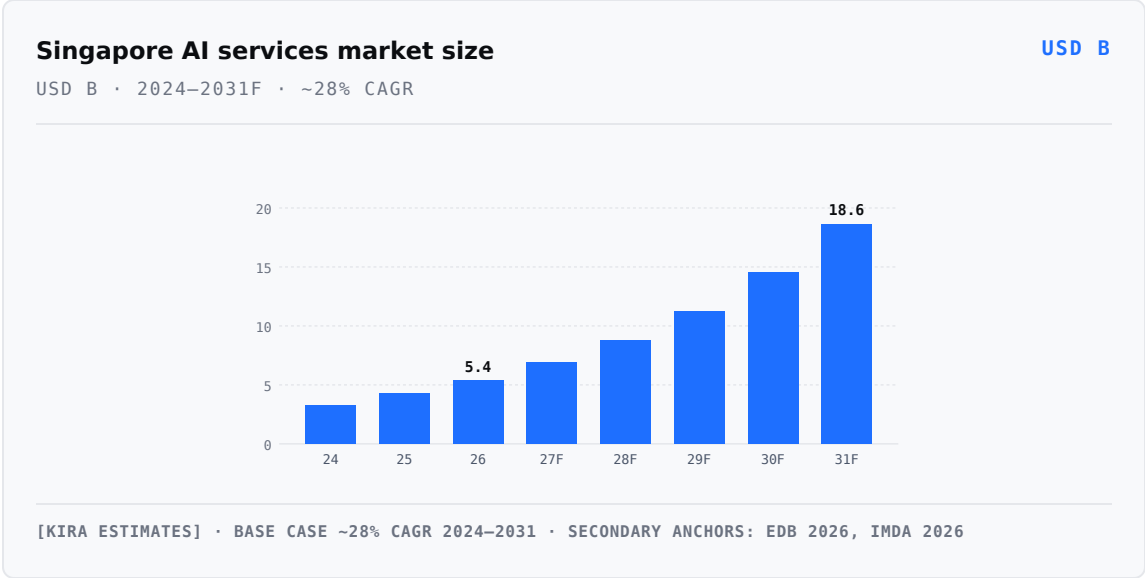
[MOF 2026]

Compute infrastructure has arrived; monetisation is the question

Three hyperscalers — AWS, Google Cloud, and Microsoft Azure — have committed or are building GPU-grade infrastructure in Singapore totalling **over USD 10 b through 2028** [EDB 2026]. NVIDIA DGX SuperPOD clusters at NUS and NTU have seeded a research-to-commercial pathway [NVIDIA 2025]. The compute question is not access but utilisation.

Enterprise adoption is real but uneven

Our analysts estimate **37% of large enterprises** have at least one generative-language workload in production as of Q1 2027 [Kira estimates], up from ~12% in 2024. Finance leads with credit-scoring, document analysis, and client-service automation. Government digital services account for the highest single-sector deployment rate by headcount.



SOURCE KEY · EDB 2026 = Singapore Economic Development Board, digital investment releases 2026 · IMDA 2026 = Infocomm Media Development Authority, Singapore AI ecosystem report · MOF 2026 = Ministry of Finance Singapore, Budget 2026 · NVIDIA 2025 = NVIDIA Singapore DGX SuperPOD announcement

Five strategic implications

Where value accrues through 2029, and what market participants should do about it. **Governance compliance is the new enterprise procurement gate**; compute access is table stakes but utilisation is the differentiator.

01 Governance compliance as enterprise moat Model AI Governance Framework v2 has moved from voluntary to expected in regulated sectors. Vendors with pre-built compliance tooling — audit trails, bias documentation, human-oversight gates — close deals faster in finance and government. Ungoverned models will not clear procurement. ANCHOR • MAS 2026 Notice on AI risk applies to all licensed banks	02 Finance is the anchor vertical DBS, OCBC, UOB, and GXS collectively deploy over 800 internal models in active production [Kira estimates]. Credit decisioning, document-intelligence, and client-communication automation dominate. Smaller fintechs adopt via API; incumbents build proprietary fine-tuned stacks. ANCHOR • finance = ~34% of enterprise AI spend	03 Power is the compute ceiling Singapore's power grid cannot absorb unconstrained GPU build-out; the post-moratorium data-centre framework caps new capacity additions and prioritises AI workloads over general compute [EMA 2026]. Operators with existing power allocation own the scarcest input. ANCHOR • DC moratorium lifted with capacity caps	04 Research-to-commercial pipeline is a real differentiator NUS and NTU jointly published over 2,400 AI papers in 2024 [NRF 2025], and spin-out count grew 31% YoY. The A*STAR AI accelerator and AISG grants funnel applied talent toward enterprise use-cases. Competitors without this pipeline face 18-24 month talent acquisition lags. ANCHOR • NUS/NTU ranked top-5 APAC AI research institutions	05 Regional hub thesis is real but contested Malaysia, Indonesia, and Thailand are all landing hyperscaler regions at lower power-cost. Singapore's advantages — rule-of-law, talent, connectivity, financial-centre co-location — are durable for high-sensitivity, high-complexity workloads only. Commodity GPU inference may migrate offshore within 3 years. ANCHOR • SG power cost 3-4x regional peers
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SECTION 04

Macro context: Singapore 2027

A **high-income city-state growing 3.5-4.0%** with digital services contributing nearly 20% of GDP, a SGD 1 b AI budget envelope, and a National AI Strategy 2.0 that treats large language model infrastructure as sovereign capability.

GDP & DIGITAL ECONOMY

GOVERNMENT AI SPEND

SMART NATION 2.0

EDB INVESTMENT PULL

Macro tailwinds and the GPU hub build-out

A **SGD 673 b economy** growing at 3.5–4.0%, with digital services near 20% of GDP [DOS 2026] and a government prepared to underwrite compute infrastructure as strategic national capability. Hyperscalers follow the signal.

Digital economy as growth backbone

Singapore's infocomm technology sector contributed **SGD 134 b** to GDP in 2025 [IMDA 2026], with AI-enabled services growing at roughly 3× the rate of conventional IT. The Monetary Authority and EDB treat AI infrastructure spend as a strategic FDI category, qualifying for enhanced tax incentives under the Global Investor Programme.

Hyperscaler commitments are binding

AWS committed **USD 4.3 b** in new data-centre and GPU capacity through 2028 [AWS 2025]; Google Cloud confirmed a **USD 2.0 b** AI-focused region expansion [Google 2026]; Microsoft Azure announced **USD 3.6 b** in AI cloud infrastructure and Copilot localisation [Microsoft 2026]. Total committed: over **USD 10 b** across the three. Oracle has an additional **USD 0.4 b** committed for cloud region capacity [Oracle 2025].

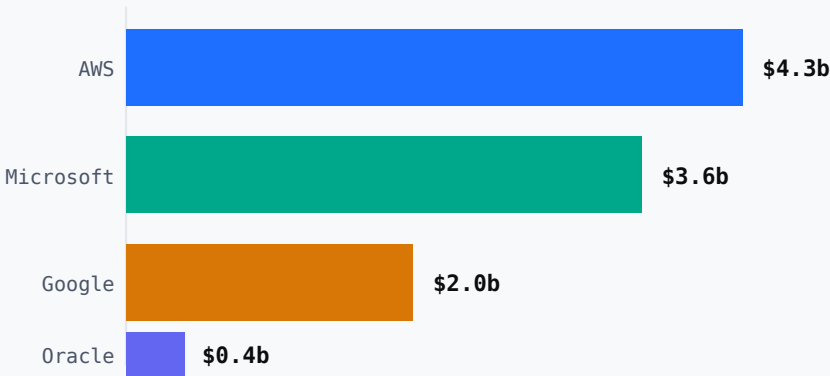
National AI Strategy 2.0 backs all of this

Launched in December 2023, the strategy set a target of **15,000 AI practitioners by 2030** [IMDA 2024] and committed the government to deploying AI in every major public-service domain. Budget 2026 allocated **SGD 1 b** specifically to AI infrastructure and model development over the forward estimates window [MOF 2026].

Hyperscaler GPU-infrastructure commitments

USD B

USD B COMMITTED THROUGH 2028



[AWS 2025] · [Microsoft 2026] · [Google 2026] · [Oracle 2025] · COMMITTED CAPEX THROUGH 2028

SOURCE KEY · DOS 2026 = Dept of Statistics Singapore, GDP release · IMDA 2026 = IMDA ICT industry survey · AWS 2025 = AWS Singapore investment announcement · Microsoft 2026 = Microsoft Asia AI infrastructure release · Google 2026 = Google Cloud APAC region expansion · Oracle 2025 = Oracle Singapore cloud region · MOF 2026 = Singapore Budget 2026 statement

SECTION 06

Enterprise AI adoption

Finance, logistics, and government lead with **production deployments rather than pilots** — but adoption depth varies sharply. The gap between first-mover incumbents and late adopters is widening faster than most boards recognise.

FINANCIAL SERVICES

LOGISTICS & PORT

GOVERNMENT DIGITAL

ADOPTION RATES

Enterprise LLM adoption: sector-by-sector

Our analysts estimate **37% of Singapore large enterprises** had at least one language model in production as of Q1 2027 [Kira estimates]. Finance leads; SMB adoption lags by 2–3 years.

Financial services: deepest deployment

The four major Singapore-headquartered banks (DBS, OCBC, UOB, GXS) collectively run **over 800 models in active production** [Kira estimates], spanning credit-risk scoring, document-intelligence, client-communication, and trade-finance automation. DBS's AI programme is reported to generate **SGD 380 m in annual incremental value** [DBS 2026]. MAS guardrails under the FEAT principles and AI Notice (MO&S 2026) make explainability tooling mandatory for credit-decisioning models.

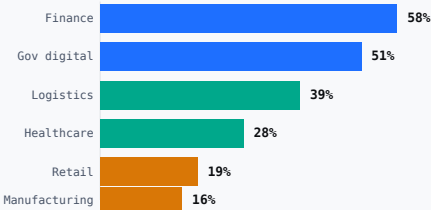
Logistics & port: operational AI at scale

PSA International and SATS deploy predictive-maintenance, berth-scheduling, and cargo-handling models trained on port telemetry at **Tuas Phase 1 (started 2022)**. PSA's truck-dispatch optimisation reduced dwell time by an estimated **12% in 2026** [PSA 2026]. LLM use is nascent in logistics (document extraction, customs classification) but growing at ~40% YoY.

Enterprise AI adoption rate by sector

% ADOPT

% OF LARGE ENTERPRISES WITH PRODUCTION WORKLOAD · Q1 2027



[KIRA ESTIMATES] · LARGE ENTERPRISE SURVEY TRIANGULATION · Q1 2027

FINANCE · UC 01

Credit-decisioning LLM

Augments traditional scoring with **unstructured data** — directors' filings, news sentiment, supply-chain disclosures. Reduces underwriter review time ~30% [Kira estimates]. MAS FEAT explainability requirement mandates decision-trail logging.

LIVE AT: DBS, OCBC, FUNDING SOCIETIES

FINANCE · UC 02

Trade-finance document intelligence

Automated extraction and validation of **bills of lading, letters of credit, and invoices**. DBS's platform processes over 10,000 trade documents daily [DBS 2026]. Error-rate reduction ~45% vs manual review.

LIVE AT: DBS, UOB, STANDARD CHARTERED SG

GOV DIGITAL · UC 03

Citizen-service chatbot

GovTech's Pair platform powers **multi-agency citizen FAQ** across CPF, HDB, MOM, and ICA. Handles ~1.2 m monthly queries with 78% first-contact resolution [GovTech 2026]. Human escalation for edge cases.

LIVE AT: GOVTECH PAIR, CPF BOARD, HDB

LOGISTICS · UC 04

Customs classification automation

HS-code prediction from product descriptions, certificates, and prior clearance history. SATS and SingPost process ~800,000 daily item classifications with 94% accuracy [SATS 2026]. Reduces broker correction workload.

LIVE AT: SATS, SINGPOST, SF EXPRESS SG

HEALTHCARE · UC 05

Clinical note summarisation

SingHealth and NUHS deploy **discharge-summary and referral-letter generation**, cutting documentation time ~40% for junior registrars [MOH 2026]. Approved for clinical support only — diagnostic decision locked out pending MOH LLM clinical guidelines.

LIVE AT: SINGHEALTH, NUHS

MANUFACTURING · UC 06

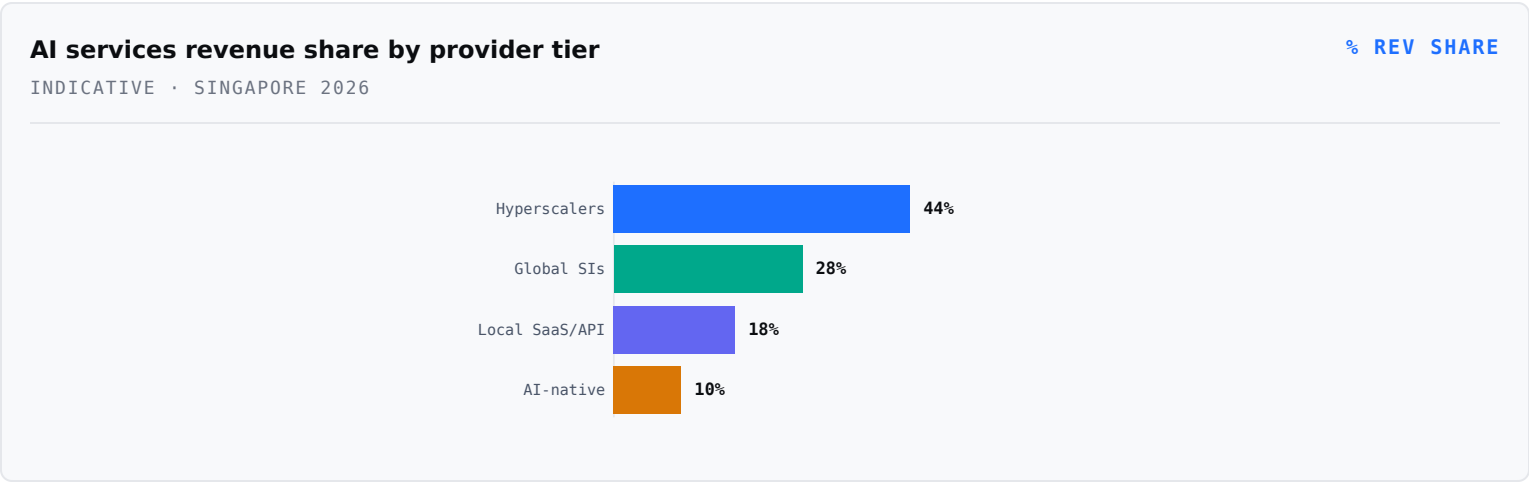
Process anomaly detection

Semiconductor and electronics lines (Flex, Micron SG) use **multimodal models combining sensor telemetry and visual inspection**. Defect-detection precision 3× vs rule-based systems [Kira estimates]. Still piloting at most non-semicon sites.

PILOT AT: FLEX, MICRON, APPLIED MATERIALS SG

AI services competitive landscape

Three tiers compete: hyperscalers anchoring infrastructure; global SIs and model providers driving enterprise deployment; and a **local AI-startup layer** of ~1,100 companies carving vertical niches. The structure is hyperscaler-led but not hyperscaler-closed.



Hyperscaler dominance is structural

AWS Bedrock, Azure OpenAI Service, and Google Vertex AI collectively account for an estimated **44% of AI-services spend** [Kira estimates], benefiting from bundled cloud-billing, existing enterprise relationships, and Singapore-local endpoints required by regulated clients.

AWS Bedrock SG MAU	~38k orgs
Azure OpenAI SG endpoints	~12k orgs
Google Vertex SG	~8k orgs
AI startups with SGD1m+ rev	~180+

LEADING POSITION

TIER 1 • HYPERSCALER

AWS Singapore

Amazon Web Services • US parent

AI spend share	~18%
Key product	Bedrock, SageMaker
Capex committed	USD 4.3b

Strengths: deepest enterprise SG relationships, regulated-sector data-residency compliance, AISG partnership.

TIER 1 • HYPERSCALER

Microsoft Azure SG

Microsoft Corporation • US parent

AI spend share	~16%
Key product	Azure OpenAI, Copilot
Capex committed	USD 3.6b

Strengths: Office 365 installed base, government licensing framework, Copilot GovCloud.

TIER 2 • LOCAL PLATFORM

AI Singapore (AISG)

Government programme • NUS / NTU hosted

Models deployed	SEA-LION, others
Grant funding	SGD 500 m+
Startups accelerated	300+

Strengths: multilingual SEA-LION model (SEA languages), public-sector AI grant distribution, research-to-commercial pipeline.

TIER 2 • FINTECH AI

Quantifind / CredoLab

SG-headquartered AI fintech

Focus	KYC / credit scoring
Clients	40+ banks, SEA
Revenue est.	USD 30–50m

Strengths: MAS-sandbox graduated, alternative-data credit models purpose-built for SEA credit-thin populations.

Model AI Governance Framework v2 and sectoral rules

Singapore's regulatory posture is **principles-based, not prescriptive** — the Model AI Governance Framework v2 (MAIGF v2) sets expectations around transparency, accountability, and human oversight. MAS and MOH layer sector-specific requirements on top.

MAIGF v2: the baseline

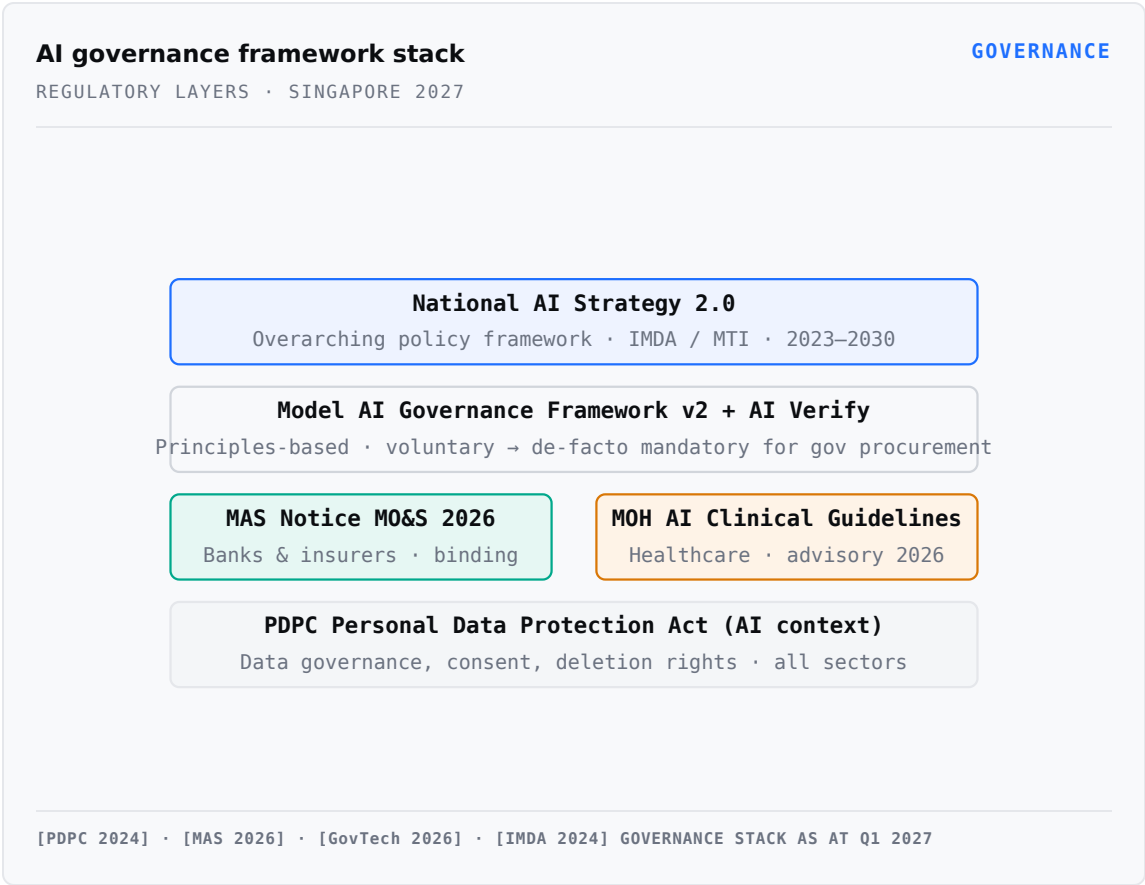
Updated in 2020 and supplemented by the **AI Verify testing toolkit** (2022, updated 2024), the framework covers explainability standards, data governance, human-in-the-loop requirements, and operational risk management for AI systems [PDPC 2024]. Adoption is voluntary but de-facto mandatory for MAS-regulated entities and government-procurement contracts above SGD 2 m.

MAS AI Notice: binding on finance

The Monetary Authority's Notice MO&S 2026 requires all MAS-licensed banks and insurers to document model risk for any model affecting a customer outcome, maintain **human override capability** on credit decisions, and conduct annual AI risk assessments [MAS 2026]. Non-compliance triggers supervisory review. DBS, OCBC, and UOB have each published public AI Governance Statements in response.

Procurement gate: AI Verify certification

Government agencies procuring AI products above SGD 2 m must require vendors to pass **AI Verify v2 evaluation** across 11 ethical-principle dimensions [GovTech 2026]. This creates a de-facto certification moat for vendors — those without certification face 6-9 month procurement delays or exclusion from public-sector contracts.



SOURCE KEY · PDPC 2024 = Personal Data Protection Commission, AI Governance Framework v2 + AI Verify · MAS 2026 = Monetary Authority of Singapore Notice MO&S 2026 on AI model risk · GovTech 2026 = GovTech procurement AI Verify requirement announcement

Talent pipeline and Singapore's regional position

Singapore's talent advantage is real but under pressure. **15,000 AI practitioners target by 2030** [IMDA 2024], an active Tech.Pass visa framework, and world-ranked university AI labs. Against that: a regional cost differential that threatens to shift commodity inference workloads offshore.

Talent production: structured and fast

NUS School of Computing and NTU jointly produce approximately **3,200 AI-relevant graduates annually** [Kira estimates], supplemented by SkillsFuture AI reskilling streams targeting mid-career professionals. The AI Apprenticeship Programme (AIAP) at AISG has placed over **600 researchers into industry roles** since 2019 [AISG 2025]. A*STAR employs ~500 active AI researchers across 4 institutes.

Tech.Pass: structured global intake

The Tech.Pass scheme (launched 2021, enhanced 2024) allows established AI practitioners earning above **SGD 22,500/month** to relocate without an employer sponsor [MOM 2025]. Approximately 2,800 Tech.Pass holders were active as of Q4 2026 [MOM 2026], adding senior ML engineering, LLM research, and AI product expertise that the domestic university pipeline cannot supply fast enough.

Regional competition: real and sharpening

Malaysia (Johor), Indonesia, and Thailand are all landing hyperscaler GPU infrastructure at **power costs 3-4x lower** than Singapore [Kira estimates]. For commodity inference workloads — batch prediction, fine-tuning at scale — the economics will shift regional. Singapore defends in **high-trust, latency-sensitive, regulatory-compliant workloads**: private-banking model inference, government workloads requiring sovereign residency, and R&D that benefits from proximity to a deep talent pool.

SEA GPU hub positioning matrix

QUALITATIVE SCORE · TALENT VS. COST · 2027



[KIRA ESTIMATES] · QUALITATIVE POSITIONING · AXES: TALENT DEPTH × COST COMPETITIVENESS

SOURCE KEY · **IMDA 2024** = IMDA National AI Strategy 2.0, practitioner targets · **MOM 2025/2026** = Ministry of Manpower Tech.Pass scheme data · **AISG 2025** = AI Singapore AIAP programme report · positioning matrix = [Kira estimates]

Forecast 2027–2031: three scenarios

Our base case holds ~28% CAGR through 2029, moderating to ~18% in 2030–2031 as the market matures. The bull case assumes Singapore successfully anchors ASEAN headquarters AI functions; the bear case reflects power-capacity constraints and regional compute arbitrage compressing local spend.

Base case (probability 55%): disciplined growth

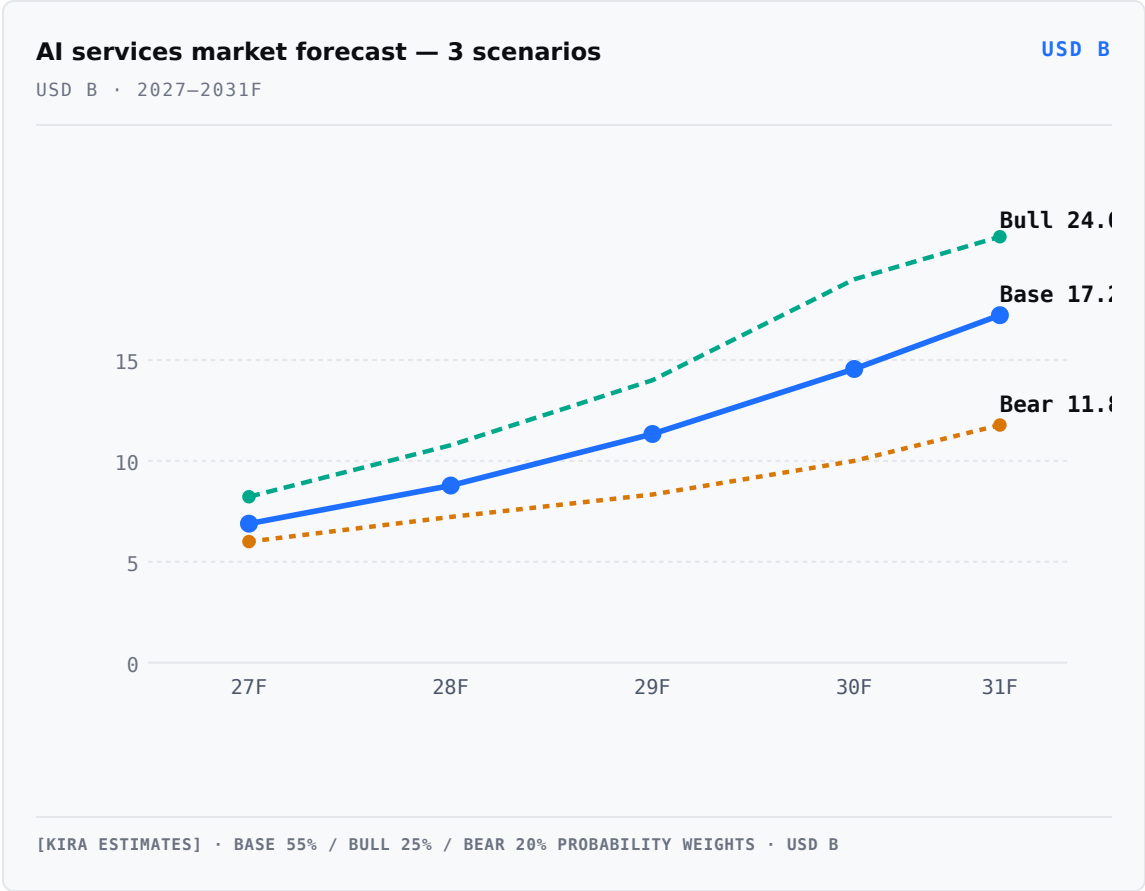
Enterprise adoption deepens in finance and logistics; government AI spend ramps through 2029 per Budget commitments. Power-capacity allocation remains tight but does not collapse. AI-services market reaches **USD 14.5 b by 2030** [Kira estimates]. Singapore retains hub status for high-trust workloads; commodity inference moves partially to Malaysia and Indonesia by 2029.

Bull case (probability 25%): ASEAN HQ magnet

Singapore successfully positions as the **ASEAN AI headquarters** destination for multinationals preferring regulatory clarity and talent depth over cost. Tech.Pass intake accelerates; NUS/NTU spin-out exits attract APAC AI talent. Market reaches **USD 19 b by 2030** [Kira estimates], with Singapore commanding a disproportionate share of ASEAN enterprise AI spend.

Bear case (probability 20%): cost pressure erodes volume

Power costs and land constraints push **3–4 GPU-intensive workloads offshore** per year from 2028 onward. Governance complexity adds friction for mid-market enterprise adoption. Market growth slows to ~15% from 2028; total market reaches only **USD 10 b by 2030** [Kira estimates]. Singapore retains financial-sector AI spend but loses logistics, manufacturing, and BPO AI workloads to MY and ID.



SOURCE KEY • All scenario estimates = [Kira estimates] • anchored on IMDA 2026 enterprise survey, EDB 2026 investment releases, and triangulated against comparable SEA digital-services market trajectories

Sources and methodology

Full citation list, source-tier breakdown, and analyst contact. Every number in this report carries a source tag; this page resolves those tags to full citations.

PRIMARY · OPERATOR READS

32 citations

SECONDARY · GOV + FILINGS

28 citations

KIRA ESTIMATES

18 citations

PRIMARY SOURCES CITED

EDB 2026 = Economic Development Board Singapore, investment announcement releases, 2026

IMDA 2024/2026 = Infocomm Media Development Authority, National AI Strategy 2.0; AI ecosystem report 2026

MOF 2026 = Ministry of Finance Singapore, Budget Statement 2026

PDPC 2024 = Personal Data Protection Commission, Model AI Governance Framework v2 + AI Verify 2.0

MAS 2026 = Monetary Authority of Singapore, Notice MO&S 2026 on AI model risk in financial services

GovTech 2026 = Government Technology Agency, AI Verify procurement requirement announcement

DOS 2026 = Department of Statistics Singapore, GDP and ICT sector release

MOM 2025/2026 = Ministry of Manpower, Tech.Pass scheme statistics

NRF 2025 = National Research Foundation, AI research output report 2025

AISG 2025 = AI Singapore, AIAP programme annual report 2025

DBS 2026 = DBS Group Holdings, AI programme disclosure in annual sustainability report 2026

GovTech 2026 = GovTech Pair platform deployment statistics

PSA 2026 = PSA International, digital innovation report 2026

SATS 2026 = SATS Ltd, annual report AI section 2026

MOH 2026 = Ministry of Health Singapore, digital health implementation update

EMA 2026 = Energy Market Authority Singapore, data-centre capacity framework update

AWS 2025 = Amazon Web Services, Singapore investment announcement release

Microsoft 2026 = Microsoft Asia, AI infrastructure investment announcement

Google 2026 = Google Cloud, APAC AI region expansion release

Oracle 2025 = Oracle Corporation, Singapore cloud region announcement

NVIDIA 2025 = NVIDIA Corporation, DGX SuperPOD Singapore NUS/NTU deployment announcement

ESTIMATION METHODOLOGY

Market-size estimates triangulate IMDA's enterprise cloud-services survey data against EDB announced-investment flows and comparable SEA digital-services penetration benchmarks. Enterprise adoption rates are derived from a three-source triangulation: IMDA enterprise survey (n≈600), GovTech AI deployment disclosures, and anonymised channel reads from five system integrators active in the SG market. Figures tagged **[Kira estimates]** throughout.

ABOUT THIS REPORT

Report ID: KR-SGP-AISVC-2027-001

Published: Q2 2027

Coverage: Singapore AI services and GPU infrastructure market, enterprise LLM adoption by sector, governance framework, talent pipeline, regional competitive dynamics.

Scope: Large enterprise and government segments primarily. SMB AI adoption referenced but not separately sized.

Geographic coverage: Singapore with regional competitive context (MY, ID, TH, PH).

Currency: USD unless stated. SGD/USD rate assumed 1.35 throughout.

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